

# Critical Slowing Down Reveals Dynamical Regimes of Slow-Moving Reservoir Landslides: Hidden Markov Model Evidence

Zhou Fang<sup>a,\*</sup>

<sup>a</sup>Business School, University of Shanghai for Science and Technology, Shanghai, 200093, China

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## ABSTRACT

Slow-moving landslides do not respond uniformly to rainfall and reservoir-level fluctuations: the same hydrological trigger can produce markedly different displacement responses depending on when it occurs. Determining whether a landslide is in a dormant or an active dynamical regime may therefore provide information complementary to conventional hydrological-threshold warning systems. This study applies critical slowing down (CSD) indicators to 17 years of monthly GPS displacement data from three slow-moving reservoir landslides in the Three Gorges Reservoir Area (Baishuihe, Bazimen, Shuping), with a two-state Gaussian hidden Markov model (HMM) serving as an independent statistical validation benchmark. Three findings emerge. First, CSD cannot provide statistically significant lead-time warning ahead of individual displacement acceleration events (all 27 permutation tests  $p > 0.05$ ). Second, CSD indicators, and Variance in particular, show consistent discrimination between dormant and active regimes across the three sites, as validated by the HMM (6/9 point-biserial correlations survive Bonferroni correction). Third, Variance consistently outperforms AR1 across all three sites (Stouffer combined  $p = 0.002$ ), a rank order that reverses the consensus of the ecological CSD literature—a reversal that may be related to the heavy-tailed distribution of monthly landslide displacement increments. Together, these findings suggest that CSD indicators carry regime-classification information not recoverable from hydrological variables alone, and contribute a nonlinear-dynamics perspective to the emerging research strand on regime identification in landslide early warning.

## 1. Introduction

The Three Gorges Reservoir Area is one of the most densely distributed regions of reservoir-induced landslides in the world. Since the impoundment of the Three Gorges Dam in 2003, the reservoir water level has been regulated on an annual cycle of approximately 12 months, fluctuating between 145 m and 175 m—drawn down before the flood season and raised afterwards—superimposed on concentrated summer monsoon rainfall from June to September. This dual cyclic hydrological forcing acts on the reservoir banks and drives the deformation of numerous slow-moving landslides (Du, Yin and Lacasse, 2013). Landslides such as Baishuihe, Bazimen, and Shuping exhibit characteristic step-like behaviour: displacement is nearly stagnant during the dry season (November to April) and accelerates during the monsoon under the combined effect of intense rainfall and rapid reservoir drawdown. The failure mode of these landslides is not sudden collapse but a gradual transition from slow creep to accelerating creep. This gradual deformation process motivates the search for indicators capable of characterising evolving system states, while imposing a distinctive requirement: such indicators must detect changes early enough to allow a response, yet with sufficient specificity to avoid the frequent false alarms that undermine operational warning systems.

Landslide displacement prediction has been a focal research area in engineering geology over the past two decades. Deep-learning methods, particularly LSTM and its variants (Conv-LSTM, Bi-LSTM, GRU), have achieved substantial gains in month-by-month displacement forecasting and have become increasingly common in the landslide literature (Gong, Zhang, Juang, Tang and Pudasaini, 2024; Ebrahim, Fares, Faris and Zayed, 2024). One noteworthy trajectory within this literature is the limited adoption of reservoir computing (RC) in subsequent landslide displacement studies. Yao, Zeng, Lian and Tang (2015); Yao, Zeng and Lian (2016) validated the predictive capacity of RC on Three Gorges landslides, but these remain the last works in the field to have used this approach; the subsequent literature

\*Corresponding author

✉ 255023292@st.usst.edu.cn (Z. Fang)

ORCID(s): 0009-0006-2309-9826 (Z. Fang)

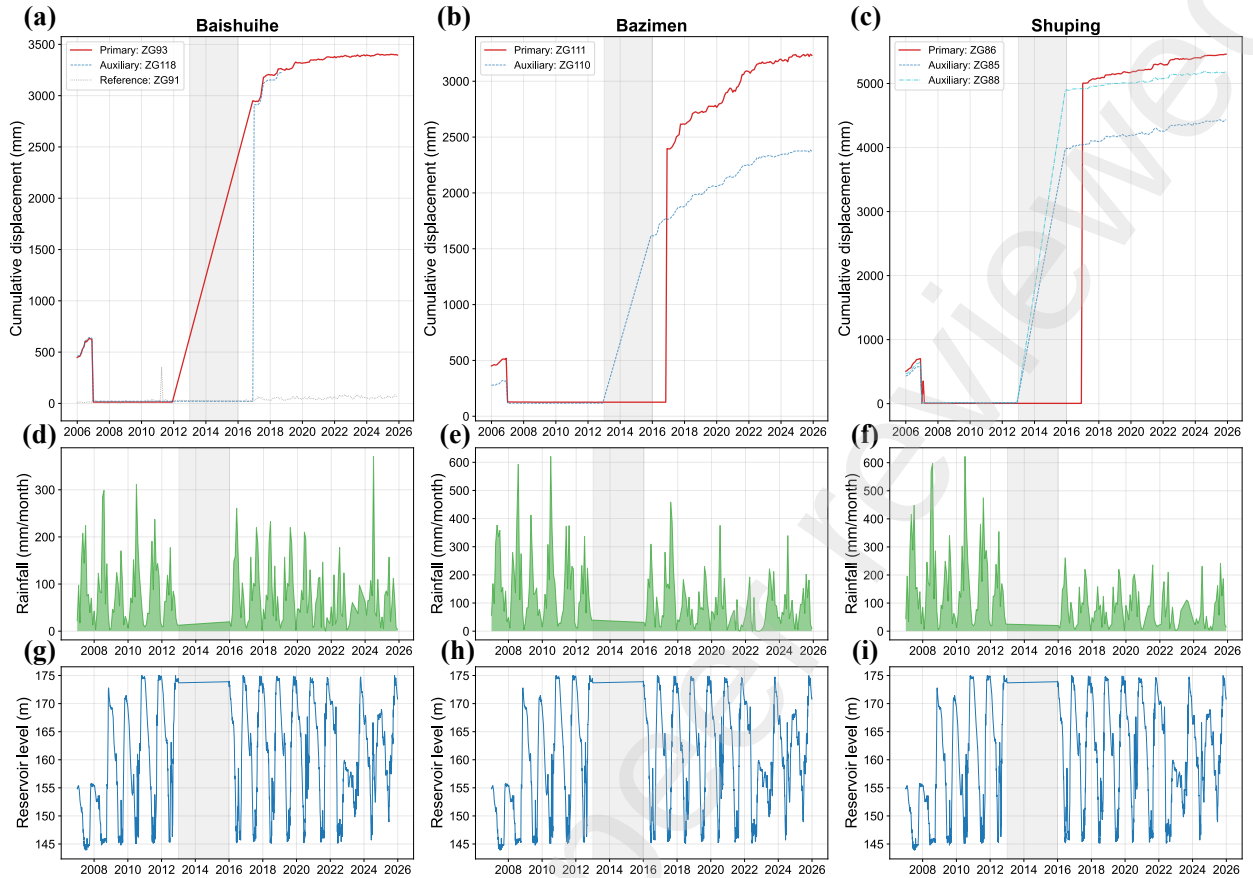
adopted deep-learning methods exclusively. Improved displacement-forecast accuracy, however, has not automatically translated into a deeper understanding of landslide dynamical states. A different question has begun to attract attention: what dynamical state is a landslide in—a “dormant regime” characterised by long quiescent intervals punctuated by occasional active episodes, or a persistently “active regime”? This question differs in kind from “what will the displacement be next month.” The former is a classification and diagnosis of the system’s current dynamical state; the latter is a point prediction of its output. The two tasks require different methodological frameworks and validation standards.

Critical slowing down (CSD) theory offers a potential physical framework for addressing the regime-classification question. As a dynamical system approaches a tipping point, its rate of recovery from small perturbations declines, a property that manifests statistically as rising autocorrelation, increasing variance, and strengthened long-range memory (Scheffer, Bascompte, Brock, Brovkin, Carpenter, Dakos, Held, van Nes, Rietkerk and Sugihara, 2009). CSD indicators, in particular the lag-1 autocorrelation coefficient  $AR(1)$  and the variance, have been extensively validated as early-warning signals of critical transitions in ecosystems (Scheffer, Carpenter, Lenton, Bascompte, Brock, Dakos, van de Koppel, van de Leemput, Levin, van Nes, Pascual and Vandermeer, 2012; Dakos, van Nes, D’Odorico and Scheffer, 2012). In the geohazard domain, CSD indicators have been applied to acoustic emission signals of rock failure (Zhang, Li, Hu, Tang and Zheng, 2021; Tang, Zhu, He, Hu and Fan, 2022), to physical-model experiments of slope failure (Yang, Fan, Yue, Chen, Wen and Zhang, 2025; Fan, Lian, Yang, Wang, Zhou, Ding and Li, 2025), and to seismic precursory signals of bedding landslides (Wang, Chen and Xu, 2024a), but their application to in-situ GPS-monitored landslide displacement remains extremely sparse. Pu, Ma, Zeng, Xu and Chen (2015) applied CSD indicators (sliding-window  $AR(1)$ , Variance, DFA) to field displacement data from the Yemaonian landslide in the Three Gorges region—the first, and to our knowledge still the only, such study—but the analysis was descriptive, limited to a single site, and lacked statistical hypothesis testing and an independent regime-validation tool. This methodological gap raises a question that goes beyond the underexplored potential of CSD in landslide monitoring: if a CSD indicator suggests a change in landslide dynamics, how can one be confident that the signal reflects a genuine dynamical state rather than a chance fluctuation in the data?

Hidden Markov models (HMMs) offer a regime-detection pathway that is independent of CSD theory. Under the CSD framework, regime classification follows a physical chain of reasoning: critical transition  $\rightarrow$  slower recovery  $\rightarrow$  rising  $AR(1)$  and Variance. HMM classification, by contrast, proceeds from purely probabilistic inference, learning a bimodal distribution from the data without invoking any hypothesis about critical transitions. The two approaches are orthogonal in their theoretical foundations. Previous applications of HMMs in the geohazard domain have concentrated on seismic-signal classification: Dammeier, Moore, Hammer, Haslinger and Loew (2016) used HMMs for automated detection of alpine rockslide signals, and Lagarde, Dietze, Hammer, Zeckra, Voigtländer, Illien, Schöpa, Hirschberg, Burtin, Hovius and Turowski (2023) used them to classify acoustic crack signals preceding a rock slope failure. In biology, Chen, Liu, Li and Chen (2016) independently demonstrated that HMMs can detect the critical state preceding phase transitions in complex biological systems, establishing a cross-disciplinary methodological link between HMMs and critical-phenomenon detection. We did not identify previous studies that apply HMM-based detection of dynamical regimes from monthly landslide displacement data for the purpose of independently validating CSD indicators.

The field of landslide early warning is undergoing a conceptual shift from prediction toward understanding. Nava, Tordesillas, Qian and Catani (2024) proposed a regime-shift detector based on a VECC residual model, identifying regime transitions in slow-moving landslides from displacement residuals rather than forecasting displacement. Wang, Diao, Guo, Wang, Guo and Li (2024b) and Du, Lian, Li, Qi, Tang, Yuan, Xu and Zeng (2025) applied decision-tree and transfer-learning methods, respectively, to the automatic classification of landslide deformation stages. The concurrent emergence of these works signals a growing collective recognition that landslide displacement is not solely a prediction problem: regime and state classification possess independent scientific value. All of these approaches, however, follow data-driven paths: ML, cointegration residuals, or decision trees. Few studies have examined landslide regime classification from the perspective of nonlinear-dynamics theory, using CSD as the characterising tool.

The present study addresses this gap through three nested questions. (1) Can CSD indicators provide statistically significant lead-time warning ahead of individual displacement acceleration events? (2) Can CSD indicators discriminate between dormant and active dynamical regimes—and, critically, does an HMM trained on displacement data alone, with no knowledge of CSD, recover the same regime partition? (3) Among the three standard CSD indicators, which one contributes most strongly to regime classification? To answer these questions, we perform a systematic CSD analysis of 17 years (2006–2012 and 2016–2025) of monthly GPS displacement data from three slow-moving reservoir landslides (Baishuihe, Bazimen, Shuping) in the Three Gorges Reservoir Area. A 2-state Gaussian HMM (Baum–Welch EM)



**Figure 1:** Landslide Monitoring 2006–2025. Columns: Baishuihe, Bazimen, Shuping. (a)–(c) Cumulative displacement at the primary (solid red) and auxiliary (dashed/dotted) GPS monitoring points. (d)–(f) Monthly rainfall. (g)–(i) Reservoir water level at the Three Gorges Dam. Grey shading: 2013–2015 data gap.

serves as an independent validation benchmark, and a multi-level statistical framework comprising permutation tests, point-biserial correlation, BIC model selection, Steiger's  $Z$ , Kendall's  $W$ , and Bonferroni multiple-testing correction is applied to all results.

## 2. Data and Study Area

### 2.1. Study area and monitoring data

The three landslides studied here—Baishuihe, Bazimen, and Shuping—are located in Zigui County, Hubei Province, along the main stream of the Yangtze River within the Three Gorges Reservoir Area (Fig. 1). All three share the same external hydrological drivers—monsoon rainfall and reservoir-level regulation—and exhibit the characteristic step-like deformation described above. Despite this shared forcing, cross-site Pearson correlations of the monthly displacement increments are weak ( $r = 0.22$ – $0.34$ ; Supplementary Fig. S7). This quasi-independence supports treating the three sites as independent lines of evidence in the subsequent cross-site validation.

Landslide displacement data were obtained from the National Cryosphere Desert Data Center (NCDC, <http://www.ncdc.ac.cn>), covering January 2006 to December 2025 at a monthly sampling frequency. The three landslides together comprise 46 GPS surface-displacement monitoring points (Baishuihe: 23; Bazimen: 12; Shuping: 11), accompanied by daily rainfall records from the Shazhenxi and Guojiaba stations in Zigui and daily reservoir water-level measurements at the Three Gorges Dam. A common data gap exists across all three sites between 2013 and 2015, during which NCDC did not make the monitoring data publicly available; this interval is marked by grey semi-transparent shading in all panels of Fig. 1. The CSD indicators are computed within each observation segment independently;

**Table 1**

Primary and auxiliary monitoring points selected for the three landslides.

| Landslide | Role                   | Point | <i>N</i> months | Coverage | $\Delta F$ (mm) | Period    |
|-----------|------------------------|-------|-----------------|----------|-----------------|-----------|
| Baishuihe | Primary                | ZG93  | 175             | 85%      | 3,393           | 2006–2025 |
|           | Auxiliary              | ZG118 | 113             | 55%      | 3,205           | 2006–2018 |
|           | Reference <sup>§</sup> | ZG91  | —               | —        | < 500           | 2006–2025 |
| Bazimen   | Primary                | ZG111 | 198             | 97%      | 3,118           | 2006–2025 |
|           | Auxiliary              | ZG110 | 198             | 97%      | 2,268           | 2006–2025 |
| Shuping   | Primary                | ZG86  | 198             | 97%      | 5,455           | 2006–2025 |
|           | Auxiliary              | ZG85  | 198             | 97%      | 4,421           | 2006–2025 |
|           | Auxiliary              | ZG88  | 198             | 97%      | 5,184           | 2006–2025 |

Note. <sup>§</sup> ZG91 is a low-displacement reference point ( $\Delta F < 500$  mm) retained as a negative control (Section 4.4). See Supplementary

Table S2 for the full list of 19 candidate points meeting the selection criteria.

sensitivity analyses confirmed that the regime classification results are consistent between the pre-2013 and post-2015 segments (Section 4.4). The 2013–2015 discontinuity in the CSD time series that results from the sliding-window computation is visible in Fig. 2 for Shuping, where it affects the AR1 and DFA panels, but it does not bias the regime-level conclusions.

Primary monitoring points were selected from the 46 available points according to three objective criteria: (1) data coverage  $\geq 50\%$  of months, to ensure statistical reliability; (2) cumulative displacement range  $\Delta F > 500$  mm, to exclude points on the landslide periphery or stable areas that function as reference benchmarks; and (3) earliest observation year  $\leq 2008$  and latest year  $\geq 2022$ , to span both the 2006–2012 and 2016–2025 data segments. These criteria yielded 19 candidates (full list in Supplementary Table S2). From these candidates, points located within the main sliding body and possessing the most complete monitoring histories were prioritised. The final primary points are ZG93 (Baishuihe), ZG111 (Bazimen), and ZG86 (Shuping). Auxiliary points were selected from the same ZG-series manual GPS system as the primary points to ensure methodological homogeneity; the consistency of their monthly displacement increments with those of the primary points is shown in Supplementary Fig. S3. ZG91 at Baishuihe is a low-displacement reference point ( $\Delta F < 500$  mm) retained as a negative control in the auxiliary-point robustness analysis (Section 4.4). Table 1 summarises the primary and auxiliary points for the three landslides; the full coverage and displacement-range data for all monitoring points are provided in Supplementary Fig. S1 and S2.

Annual displacement increment totals for each primary point are provided in Supplementary Fig. S4. The analysis variable is the monthly displacement increment  $\delta(t) = F(t) - F(t-1)$ , where  $F(t)$  denotes the cumulative resultant horizontal displacement recorded by the GPS point in the given month.  $\delta(t)$  is chosen in preference to  $F(t)$  because cumulative displacement series are typically non-stationary (they contain a long-term creep trend), whereas the first difference removes this trend component, yielding a statistically more stationary series (Section 2.2) while preserving the amplitude information of monsoon-step events; this amplitude information is essential for the regime-discriminating capacity of the CSD indicators.

Following the standard landslide activity classification of Cruden and Varnes (1996), we refer to the two kinematic states observable in  $\delta(t)$  as the **dormant regime** (months with near-zero displacement, typically the dry season from November to April) and the **active regime** (months with measurable displacement, concentrated in the monsoon season from June to September). The core aim of the present study is to determine whether CSD indicators can discriminate these two regimes from  $\delta(t)$  alone, and whether an independent HMM—trained on the same data but with no knowledge of CSD theory—recovers the same regime partition.

## 2.2. Statistical properties of monthly displacement increments

The stationarity properties and distributional characteristics of  $\delta(t)$  at the three primary points are summarised in Table 2. The  $\delta(t)$  series for Baishuihe and Shuping pass the dual ADF and KPSS stationarity tests (ADF  $p < 0.01$ ; KPSS  $p > 0.05$ ). The  $\delta(t)$  series for Bazimen is formally non-stationary under this joint criterion (ADF  $p = 0.26$ ; KPSS  $p = 0.02$ ). Several considerations mitigate this finding. First, the CSD indicators are computed within local sliding windows ( $W = 24$  months, Section 3.2), within which approximate stationarity can be assumed. Second, the 2-state HMM is inherently suited to multi-regime distributions: each regime possesses its own mean and variance and

**Table 2**

Stationarity and distributional properties of the monthly displacement increment  $\delta(t)$  at the three primary monitoring points.

| Landslide | Stationarity            |         |          | Distribution     |          |                 |
|-----------|-------------------------|---------|----------|------------------|----------|-----------------|
|           | $\delta(t)$ stationary? | ADF $p$ | KPSS $p$ | Shapiro–Wilk $p$ | Skewness | Excess kurtosis |
| Baishuihe | Yes                     | <0.01   | 0.10     | <0.001           | +3.94    | +19.57          |
| Bazimen   | No                      | 0.26    | 0.02     | <0.001           | +1.50    | +5.38           |
| Shuping   | Yes                     | <0.01   | 0.09     | <0.001           | +1.67    | +4.34           |

*Note.* Stationarity assessed via the dual ADF + KPSS criterion: ADF  $p < 0.05$  rejects the null of a unit root; KPSS  $p > 0.05$  fails to reject the null of stationarity. “Yes” indicates that both conditions are satisfied.

does not require global stationarity. Most importantly, Bazimen produced the strongest CSD–HMM cross-validation result among the three sites (Variance  $r_{pb} = 0.757$ , Section 4.2), indicating that the regime signal dominates any weak non-stationarity in the series. This limitation is acknowledged in the Discussion.

Shapiro–Wilk tests reject normality for  $\delta(t)$  at all three sites ( $p < 0.001$ ). The distributions are positively skewed (skewness +1.50 to +3.94) and exhibit pronounced excess kurtosis (+4.34 to +19.57); the extreme value at Baishuihe (+19.57) is driven largely by a single anomalous monsoon month in July 2017 ( $\delta \approx 95$  mm). This heavy-tailed character bears directly on the central finding of the paper: Variance is highly sensitive to extreme monthly displacements (a single month can sharply increase the sliding-window variance), whereas AR(1), a Pearson correlation coefficient that depends only on the rank ordering of data points rather than their magnitude, is nearly immune to such outliers. This asymmetry constitutes the physical mechanism underlying the result that Variance significantly outperforms AR(1) (Section 4.3), rather than it being a mere statistical artefact. Quantile–quantile plots of  $\delta(t)$  are provided in Supplementary Fig. S6.

The lagged influence of rainfall and reservoir water level on  $\delta(t)$  was assessed via cross-correlation functions (CCF) and Granger causality tests (full CCF table and Granger results in Supplementary Table S3). Within the physically constrained lag range of 1–6 months, recommended lags satisfying  $r > 0$  (physically meaningful direction) and  $p < 0.05$  (statistical significance) are as follows. For rainfall: 0–2 months (Baishuihe: lag 1,  $r = +0.21$ ,  $p = 0.03$ ; Bazimen: lag 1,  $r = +0.29$ ,  $p = 0.004$ ; no significant lag at Shuping). For reservoir water level: 4–6 months (Bazimen: lag 6,  $r = +0.39$ ,  $p < 0.001$ ; Shuping: lag 4,  $r = +0.40$ ,  $p < 0.001$ ; no significant lag at Baishuihe). The lag structure is displayed as CCF stem plots in Supplementary Fig. S5. Notably, both Granger tests are non-significant at Baishuihe (rainfall  $p = 0.32$ ; water level  $p = 0.50$ ), indicating that the autoregressive component of  $\delta(t)$  already captures most of the variance at this site, a pattern consistent with a non-linear, path-dependent coupling between hydrological forcing and landslide kinematic response that is not fully detectable within a linear Granger framework. This lag structure provides the quantitative basis for the hydrological cross-analysis in Section 4.2 and for the physical interpretation, developed in the Discussion, of why CSD classifies dynamical regimes yet fails at event-level prediction.

### 3. Methods

The analytical framework proceeds in four sequential steps: critical slowing down (CSD) indicators are extracted from the monthly displacement increment  $\delta(t)$  of each landslide (Section 3.1); a sliding-window parameterisation is established to determine the temporal resolution of these indicators (Section 3.2); a hidden Markov model (HMM) is employed as an independent detector of the dynamical regimes that the CSD indicators are designed to capture (Section 3.3); finally, a suite of statistical procedures is applied to validate the full set of results (Section 3.4).

#### 3.1. Critical Slowing Down indicators

Critical slowing down (CSD) is a generic property of dynamical systems approaching a tipping point: the rate at which the system recovers from external perturbations declines, and this slowing manifests statistically as rising autocorrelation and increasing variance (Scheffer et al., 2009). Three standard CSD indicators are computed from the monthly displacement increment  $\delta(t)$  of each landslide.

**Lag-1 autocorrelation coefficient, AR(1):**  $AR(1)_t = \text{corr}(\delta_{t-W+1:t-1}, \delta_{t-W+2:t})$ , the Pearson correlation between adjacent values of  $\delta$  within the sliding window. As AR(1) approaches unity, successive monthly displacement increments become increasingly similar: the system's "memory" lengthens and its recovery from perturbations slows.

**Variance:**  $\text{Var}_t = \frac{1}{W-1} \sum_{i=t-W+1}^t (\delta_i - \bar{\delta})^2$ , the sample variance of  $\delta$  within the sliding window. Near a critical transition, the amplitude of the system's response to external forcing may increase markedly even if the forcing intensity remains unchanged.

**Detrended Fluctuation Analysis exponent, DFA:** the cumulative deviation series of  $\delta$  within the sliding window is analysed across multiple time scales; the DFA exponent  $\alpha$  is the scaling exponent in  $\log F(s) \sim \alpha \log s$ . Values of  $\alpha$  approaching unity indicate strengthened long-range memory, providing a supplementary signal of critical dynamics.

These three indicators characterise the same dynamical system from complementary statistical dimensions: AR(1) captures temporal dependence, Variance captures fluctuation amplitude, and DFA captures long-range memory. AR(1) and Variance are the most widely used indicators in the CSD literature (Dakos et al., 2012); DFA is included as a supplementary metric (Lenton, Livina, Dakos, van Nes and Scheffer, 2012). In the landslide context, the monthly displacement increment  $\delta(t)$  represents the deformation response to the combined hydrological perturbation of rainfall and reservoir-level change; a transition from dormant to active regime is accompanied by a systematic change in the amplitude and temporal-dependence structure of this response, which is the type of statistical change that CSD indicators are designed to detect. This study applies CSD indicators to field-monitored GPS landslide displacement to characterise the dormant/active dynamical regimes of three slow-moving reservoir landslides in the Three Gorges Reservoir area.

### 3.2. Sliding window parameterization

The three CSD indicators are computed within a sliding window rather than over the full  $\delta(t)$  series. The window width  $W$  governs the trade-off between temporal resolution and statistical reliability. The reference window is set to  $W = 24$  months with a step of 1 month. The physical rationale for  $W = 24$  is that the hydrological forcing of the Three Gorges landslides operates on an annual cycle of approximately 12 months (the summer monsoon season combined with seasonal reservoir-level regulation); a 24-month window spans two complete hydrological cycles, filtering out intra-annual seasonal fluctuations while preserving the low-frequency signal of cross-annual regime transitions. The robustness of this choice is assessed in Section 4.4 through a sensitivity analysis over  $W \in \{12, 18, 24, 30, 36\}$  months.

### 3.3. Hidden Markov Model for regime detection

Whether the CSD indicators genuinely capture the dynamical regimes of the landslides requires a regime detector that is **independent of CSD theory**. A two-state Gaussian hidden Markov model (2-state Gaussian HMM) is adopted as an orthogonal cross-validation benchmark (Rabiner, 1989). The HMM assumes that the observed sequence  $\delta(t)$  is generated by an unobserved binary regime variable  $S_t \in \{0, 1\}$  (0 = dormant, 1 = active) evolving as a first-order Markov process, with each regime associated with an independent Gaussian distribution  $\mathcal{N}(\mu_k, \sigma_k^2)$ ,  $k \in \{0, 1\}$ . The model parameters—the initial state distribution  $\pi$ , the state transition matrix  $\mathbf{A}$ , and the emission means and variances  $\{\mu_k, \sigma_k^2\}$ —are estimated from  $\delta(t)$  via maximum likelihood using the Baum–Welch expectation–maximisation (EM) algorithm (Bilmes, 1998), with 200 iterations and a convergence tolerance of  $10^{-4}$ . Model selection between the 2-state and 1-state specifications is guided by the Bayesian Information Criterion (Section 4.2). The two HMM states are interpreted as corresponding, respectively, to the dormant and active dynamical regimes of the landslides. This interpretation is made post hoc—the HMM was trained on  $\delta(t)$  alone and was not informed of the regime labels; the assignment of the low-mean state to “dormant” and the high-mean state to “active” follows from the mean absolute displacement in each state (Table 4). The HMM receives only  $\delta(t)$  as input; it has no access to the CSD indicators and therefore provides independent statistical validation of the regime-classification capacity of the CSD framework. The HMM is implemented in pure Python (class `TwoStateHMM`); the Baum–Welch EM initialisation uses the sample median  $\pm$  one standard deviation as the initial guesses for the two regime means. The results are identical across five random seeds (Section 4.4), confirming that the learned regime structure is deterministic rather than an artefact of random initialisation.

### 3.4. Statistical validation framework

Six statistical procedures form a multi-level validation framework.

**Permutation test.** The dates of acceleration events are randomly shuffled 10,000 times to construct a null distribution of the CSD hit rate; the one-tailed  $p$ -value of the observed hit rate is computed against this distribution.

This test provides the formal statistical answer to whether CSD can deliver significant lead-time warning ahead of individual events (Section 4.1).

**Point-biserial correlation ( $r_{pb}$ ).** The binary HMM regime labels are correlated with the CSD indicator values via the point-biserial coefficient. A significantly positive  $r_{pb}$  indicates that the CSD indicator is systematically higher during active months than during dormant months. This test serves the CSD–HMM cross-validation in Section 4.2.

**BIC model selection.**  $BIC = -2 \ln \mathcal{L} + k \ln N$  compares the goodness-of-fit of the 1-state and 2-state HMM specifications to  $\delta(t)$  (Kass and Raftery, 1995). A difference  $\Delta BIC = BIC_{1st} - BIC_{2st} > 10$  is taken as strong evidence favouring the 2-state model (Kass and Raftery, 1995). This test validates the HMM model order in Section 4.2.

**Steiger's  $Z$  test.** The point-biserial correlations of two CSD indicators (Variance vs. AR1) with the same set of HMM regime labels are compared for a statistically significant difference (Steiger, 1980). The results from the three sites are combined via Stouffer's method for cross-site meta-analysis (Koçak, 2016). This test supplies the formal statistical answer to whether Variance significantly outperforms AR1 (Section 4.3).

**Kendall's  $W$  cross-site concordance.** The rankings of the CSD–HMM point-biserial correlations across the three sites are tested for cross-site agreement (Kendall and Babington Smith, 1939). Note that Kendall's  $W$  (coefficient of concordance) used here is distinct from Kendall's  $\tau$ , which is employed for trend testing elsewhere in the early-warning-signals literature (Chen, Ghadami and Epureanu, 2022). This test evaluates cross-site consistency in Section 4.4.

**Bonferroni multiple-testing correction.** All 18 significance tests (9 permutation tests from Section 4.1 plus 9 point-biserial correlation tests from Section 4.2) are subjected to a Bonferroni correction, controlling the family-wise error rate at  $\alpha_{adj} = 0.05/18 \approx 0.0028$ .

## 4. Results

### 4.1. CSD cannot predict individual acceleration events

Before examining whether CSD indicators can discriminate between dynamical regimes, a more direct question must be addressed: can these indicators provide statistically significant lead-time warning ahead of individual displacement acceleration events? An acceleration event is defined as a month in which  $\delta(t)$  exceeds the 95th percentile for that site (the full event list appears in Supplementary Table S4). Under this definition, 4 events are identified at Baishuihe ( $\delta > 22$  mm), 9 at Bazimen ( $\delta > 28$  mm), and 7 at Shuping ( $\delta > 21$  mm). The variation in event counts follows from differences in the shape of the  $\delta(t)$  distributions. Baishuihe contains an extreme outlier (July 2017,  $\delta = 95$  mm), which substantially elevates the 95th percentile and reduces the number of months meeting the threshold. At Bazimen, moderately large  $\delta$  values are more evenly distributed than at the other two sites, yielding a lower 95th percentile and hence more identified events. Each of these thresholds follows directly from the percentile computation and involves no subjective tuning.

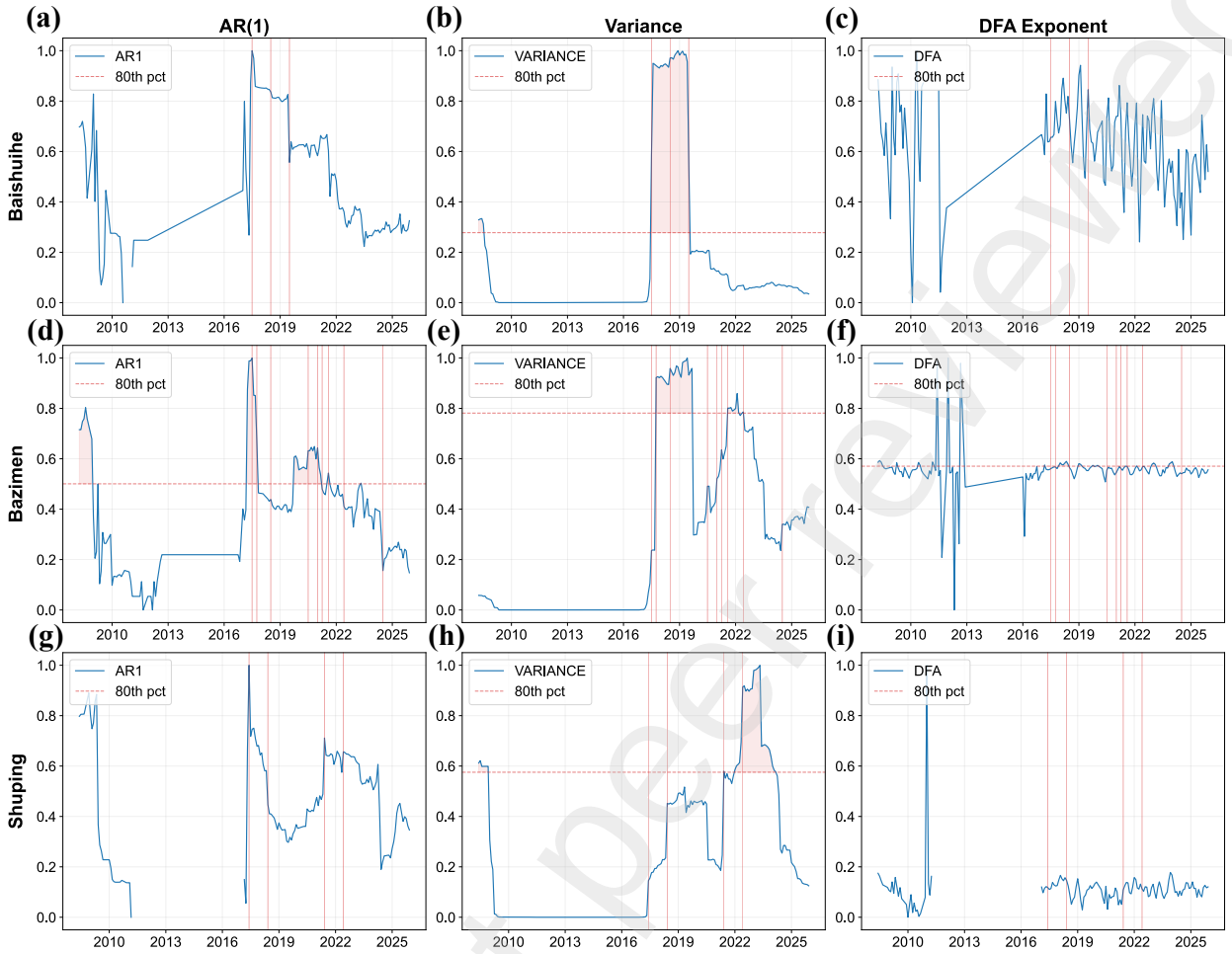
These events were then compared against the temporal profiles of the three CSD indicators, computed within overlapping 24-month sliding windows and normalised to  $[0, 1]$  (Fig. 2). Normalisation is essential because the raw units—a correlation coefficient (AR1), squared millimetres (Variance), and a dimensionless scaling exponent (DFA)—are mutually incommensurable; without it, no common visual scale exists for comparing their temporal patterns. Were CSD indicators capable of providing advance warning, the months in which an indicator exceeds its 80th-percentile threshold (red shading) would systematically precede the acceleration events (red vertical lines).

The observed pattern points to the contrary. Red shading is visually discernible only in the Variance panels; in the remaining four panels—Baishuihe AR1 and DFA, Shuping AR1 and DFA—the 80th-percentile reference line lies at the lower axis boundary, reflecting the near-constancy of CSD values at these site–indicator combinations.

Visual impression, however, is insufficient to establish a null result. Formal statistical evidence was obtained through the permutation test described in Section 3.4: event dates were randomly shuffled 10,000 times, and the one-tailed  $p$ -value of the observed hit rate was computed against the resulting null distribution.

All nine permutation  $p$ -values exceed 0.05 (Table 3). At Baishuihe, the observed hit rates are 0.75 for AR1 ( $p = 0.65$ ), 0.75 for Variance ( $p = 0.68$ ), and 0.50 for DFA ( $p = 0.82$ ). At Bazimen, the corresponding values are 0.89 for AR1 ( $p = 0.85$ ), 0.89 for Variance ( $p = 0.88$ ), and 0.78 for DFA ( $p = 0.92$ ). At Shuping, the hit rates are 0.43 for AR1 ( $p = 0.94$ ), 0.29 for Variance ( $p = 0.98$ ), and 0.43 for DFA ( $p = 0.96$ ). Even at the conventional  $\alpha = 0.05$  level, none of the nine tests rejects the null hypothesis that the CSD hit rate is no better than random.

This null result is not an artefact of the 95th-percentile threshold. Repeating the full suite of 27 permutation tests (3 event thresholds  $\times$  3 sites  $\times$  3 indicators) with the threshold relaxed to the 90th and 85th percentiles yields  $p$ -values that remain uniformly above 0.05 (Supplementary Table S5; see also Section 4.4). The conclusion is robust: CSD



**Figure 2:** CSD Indicators and Acceleration Events. Three critical slowing down indicators normalised to  $[0, 1]$  are overlaid with acceleration events (red vertical lines;  $\delta(r)$  exceeding the 95th percentile). Rows: Baishuihe (a)–(c), Bazimen (d)–(f), Shuping (g)–(i). Columns: AR1 (a,d,g), Variance (b,e,h), DFA (c,f,i). The dashed red line marks the 80th-percentile threshold; red shading indicates months in which the indicator exceeds this threshold.

indicators cannot provide statistically significant lead-time warning ahead of individual acceleration events, regardless of how generously those events are defined.

The question that follows is whether these same indicators can serve a different purpose—classifying the underlying dynamical regime in which a landslide is operating, rather than predicting the timing of individual displacement accelerations.

#### 4.2. CSD-HMM cross-validation confirms regime classification

The question posed at the end of Section 4.1 is whether CSD indicators can classify the dormant and active dynamical regimes that the HMM learns from  $\delta(r)$  alone. Because the HMM is trained entirely independently of the CSD indicators (Section 3.3), any significant association between the two constitutes independent cross-validation evidence.

The regime classification produced by the HMM for each landslide is displayed in Fig. 3. At all three sites, the HMM recovers a binary structure from the displacement increment data alone that aligns closely with physical expectation. Blue markers (dormant months) cluster tightly around  $\delta \approx 0$ ; grey markers (active months) scatter across a wide range of positive and negative values, corresponding to the displacement surges and partial rebounds driven by monsoon rainfall and reservoir drawdown. The HMM received no external labels, no seasonal calendar, and no hydrological

**Table 3**

Permutation test of CSD lead-time warning ahead of individual acceleration events.

| Site      | Events<br>( <i>N</i> ) | AR1      |                   | Variance |                   | DFA      |                   |
|-----------|------------------------|----------|-------------------|----------|-------------------|----------|-------------------|
|           |                        | Hit rate | $p_{\text{perm}}$ | Hit rate | $p_{\text{perm}}$ | Hit rate | $p_{\text{perm}}$ |
| Baishuihe | 4                      | 0.75     | 0.65              | 0.75     | 0.68              | 0.50     | 0.82              |
| Bazimen   | 9                      | 0.89     | 0.85              | 0.89     | 0.88              | 0.78     | 0.92              |
| Shuping   | 7                      | 0.43     | 0.94              | 0.29     | 0.98              | 0.43     | 0.96              |

*Note.* Hit rate: proportion of events for which at least one month within the preceding 1–6 months shows a CSD indicator exceeding the 80th-percentile threshold.  $p$ -values are one-tailed, derived from 10,000 random shuffles of event dates.

**Table 4**

HMM regime diagnostics and model selection.

| Site      | Regime diagnostics             |                               |           | Transition persistence |          |
|-----------|--------------------------------|-------------------------------|-----------|------------------------|----------|
|           | mean $ \delta $ , dormant (mm) | mean $ \delta $ , active (mm) | Dormant % | $P_{00}$               | $P_{11}$ |
| Baishuihe | 1.0                            | 10.8                          | 83%       | 0.957                  | —        |
| Bazimen   | 0.2                            | 12.3                          | 37%       | 0.985                  | —        |
| Shuping   | 0.0                            | 8.3                           | 34%       | 0.865                  | —        |

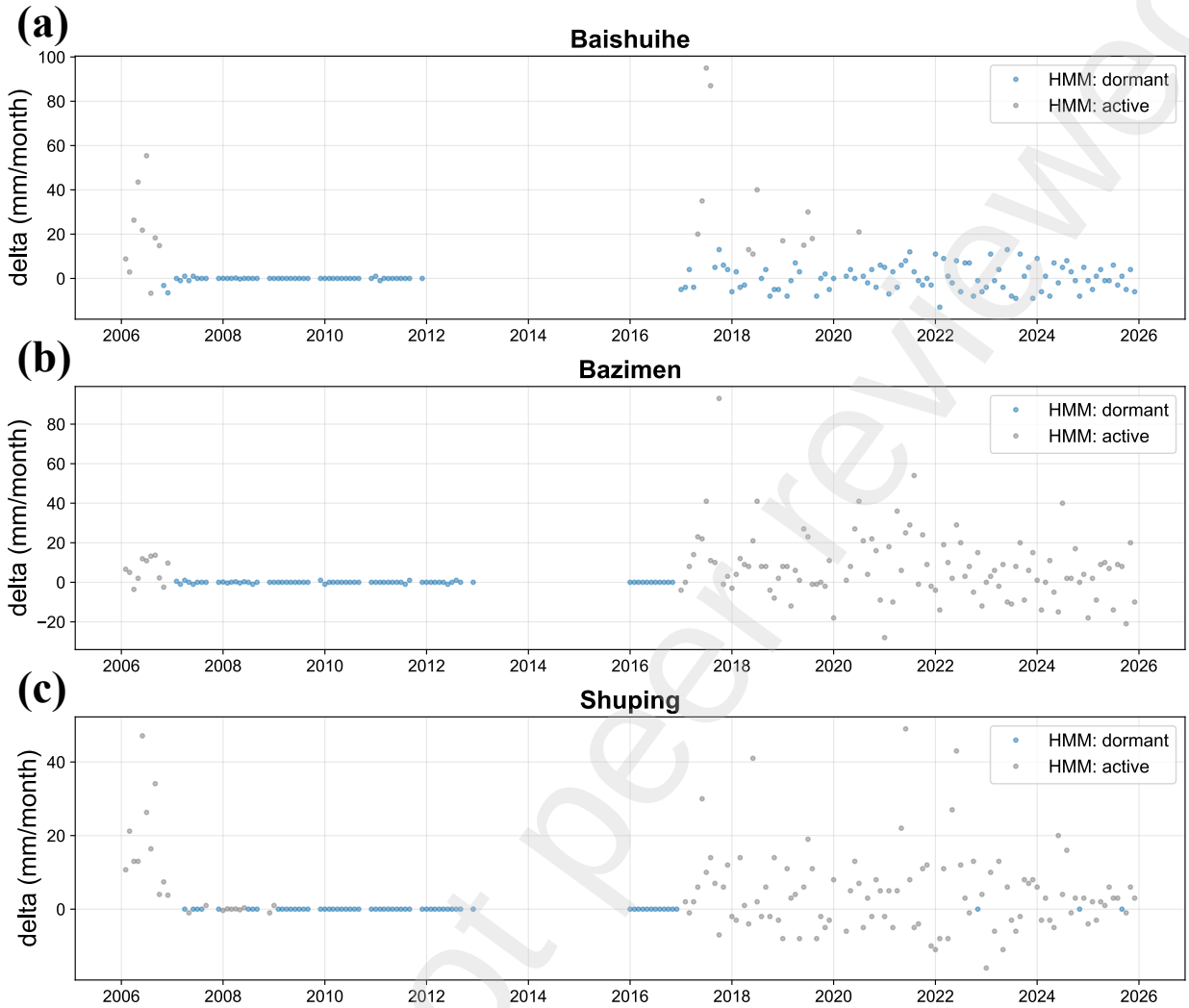
| BIC model selection |              |              |                    |
|---------------------|--------------|--------------|--------------------|
| Site                | BIC(1-state) | BIC(2-state) | $\Delta\text{BIC}$ |
| Baishuihe           | 1344.3       | 1086.4       | +257.9             |
| Bazimen             | 1506.2       | 1079.6       | +426.6             |
| Shuping             | 1369.2       | 234.0        | +1135.2            |

*Note.*  $P_{00} = P(\text{dormant} \rightarrow \text{dormant})$ , the diagonal transition probability.  $\Delta\text{BIC} = \text{BIC}_{1\text{st}} - \text{BIC}_{2\text{st}}$ ;  $\Delta\text{BIC} > 10$  indicates strong evidence for the 2-state model (Kass and Raftery, 1995). Regimes learned by Baum–Welch EM (200 iterations, identical across 5 random seeds).

data: the Baum–Welch EM algorithm converged to this bimodal structure purely from the  $\delta(t)$  distribution, and the result is identical across five random seeds (Section 4.4).

Quantitative diagnostics of the two regimes are provided in the upper panel of Table 4. Across all three landslides, the mean absolute displacement during dormant months is 0.0–1.0 mm: the landslides are essentially stationary during these periods. In active months, by contrast, mean  $|\delta|$  rises to 8.3–12.3 mm, an order of magnitude larger. The diagonal elements of the transition matrix are close to unity ( $P_{00} = 0.865$ – $0.985$ ), indicating high persistence within each regime: once a landslide enters either state, it is likely to remain there for several consecutive months. The proportion of dormant months differs markedly across sites: 83% at Baishuihe versus 37% at Bazimen and 34% at Shuping. This contrast is consistent with the visual impression from Fig. 1—at Baishuihe, long quiescent intervals are punctuated by infrequent active episodes, whereas at Bazimen and Shuping the active periods are more continuous and frequent.

Having established that the HMM identifies physically meaningful regimes, the next step is to test whether the CSD indicators can recover the same partition. Point-biserial correlations between the three CSD indicators and the binary HMM regime labels are reported in Table 5. Seven of the nine tests are nominally significant at  $\alpha = 0.05$ : CSD indicators are systematically higher during active months than during dormant months. At Baishuihe, AR1 ( $r_{\text{pb}} = +0.362$ ) and Variance ( $r_{\text{pb}} = +0.354$ ) perform comparably. At Bazimen, Variance yields the strongest cross-validation signal in the entire study ( $r_{\text{pb}} = +0.757$ ). At Shuping, Variance again outperforms the other two indicators ( $r_{\text{pb}} = +0.611$ ). DFA is non-significant at both Bazimen ( $r_{\text{pb}} = +0.051$ ) and Shuping ( $r_{\text{pb}} = -0.043$ ), consistent with the pattern observed throughout: DFA is the weakest of the three CSD indicators. Under the conservative Bonferroni correction ( $\alpha_{\text{adj}} = 0.05/18 \approx 0.0028$ ), six of the nine correlations survive: AR1 and Variance remain significant at all three sites. The regime-classification capacity of the CSD framework therefore holds across all sites and both core indicators, even under the most stringent statistical safeguard.



**Figure 3: HMM 2-State Regime Detection.** Monthly displacement increment  $\delta(t)$  at the three primary monitoring points, coloured by HMM-inferred regime. Blue: dormant months (mean  $|\delta| \approx 0\text{--}1$  mm). Grey: active months (mean  $|\delta| \approx 5\text{--}15$  mm). The HMM was trained on  $\delta(t)$  alone with no external regime labels.

The choice of a 2-state HMM, rather than a simpler 1-state alternative, is independently supported by the Bayesian Information Criterion (lower panel of Table 4). The differences  $\Delta\text{BIC} = \text{BIC}_{1\text{st}} - \text{BIC}_{2\text{st}}$  are +257.9 (Baishuihe), +426.6 (Bazimen), and +1135.2 (Shuping), all far exceeding the conventional threshold of  $\Delta\text{BIC} > 10$  for strong evidence (Kass and Raftery, 1995). The  $\delta(t)$  series are drawn from two distributions, not one; the 2-state HMM is not an arbitrary modelling choice but the model order that the data themselves support.

A final methodological check addresses whether the HMM regime labels might merely function as a statistical proxy for rainfall intensity or water-level fluctuations. Mann–Whitney  $U$  tests and logistic regression argue against this interpretation. At two of the three sites (Bazimen and Shuping), antecedent rainfall is higher during dormant months than during active months—the opposite of the intuitive expectation. The McFadden  $R^2$  of logistic regressions predicting HMM state from lagged rainfall and water-level change ranges from 0.035 to 0.134 (mean 0.079): hydrological variables carry almost no predictive power for the HMM regime labels. Complete stratification statistics and logistic regression coefficients are provided in Supplementary Table S11.

The CSD–HMM cross-validation yields a consistent positive result across three independent landslides. Both core CSD indicators, AR1 and Variance, remain significantly correlated with the HMM regime partition after Bonferroni

**Table 5**

Point-biserial correlation between CSD indicators and HMM regime labels.

| Site      | AR1      |                               | Variance      |                               | DFA      |                       |
|-----------|----------|-------------------------------|---------------|-------------------------------|----------|-----------------------|
|           | $r_{pb}$ | $p$                           | $r_{pb}$      | $p$                           | $r_{pb}$ | $p$                   |
| Baishuihe | +0.362   | $1.10 \times 10^{-5\dagger}$  | +0.354        | $1.48 \times 10^{-5\dagger}$  | +0.189   | $3.88 \times 10^{-2}$ |
| Bazimen   | +0.492   | $2.42 \times 10^{-11\dagger}$ | <b>+0.757</b> | $2.73 \times 10^{-32\dagger}$ | +0.051   | $5.34 \times 10^{-1}$ |
| Shuping   | +0.261   | $2.48 \times 10^{-3\dagger}$  | +0.611        | $6.26 \times 10^{-18\dagger}$ | -0.043   | $6.25 \times 10^{-1}$ |

Note.  $\dagger$  Survives Bonferroni correction at  $\alpha_{adj} = 0.05/18 \approx 0.0028$ . HMM regime labels: 0 = dormant, 1 = active. A positive  $r_{pb}$

indicates that the CSD indicator is systematically higher during active months.

correction; the BIC independently supports the 2-state specification; and the HMM regime labels cannot be trivially predicted from external hydrological variables. These converging results establish that CSD indicators carry regime-classification information that cannot be recovered from hydrological variables alone. The following section turns to a more specific comparison: among the three CSD indicators, which one contributes most strongly to regime classification?

#### 4.3. Variance outperforms AR(1) as a regime indicator

Section 4.2 demonstrated that CSD indicators—AR1 and Variance in particular—can distinguish the HMM-learned dormant and active regimes. A natural question follows: between these two indicators, which one is the more reliable classifier?

The per-site evidence in Table 5 already hints at an answer. At Bazimen, the Variance  $r_{pb}$  (+0.757) is nearly 1.5 times that of AR1 (+0.492); at Shuping the gap is comparably large (+0.611 vs. +0.261); at Baishuihe the two are nearly equal (+0.354 vs. +0.362). Visual comparison of two correlation magnitudes, however, is not statistical evidence. A formal comparison requires Steiger's  $Z$  test for dependent correlations (Steiger, 1980), applied to the two point-biserial coefficients computed against the same HMM regime labels, followed by Stouffer's method for cross-site meta-analysis (Kocak, 2016). Table 6 reports the results.

Bazimen is the only site with sufficient statistical power to resolve the comparison formally: Variance significantly outperforms AR1 at this site (Steiger  $Z = +4.99$ ,  $p < 0.001$ ). At Baishuihe and Shuping the two indicators produce similar point-biserial values and the individual-site differences do not reach significance. Combining the three sites via Stouffer's method, the cross-site meta-analysis supports the conclusion that Variance outperforms AR1 (combined  $Z = +2.88$ , one-tailed  $p = 0.002$ ). Neither the variance–DFA nor the AR1–DFA comparison reaches significance at either the per-site or the meta-analytic level, reinforcing the pattern that DFA is the weakest of the three indicators throughout this study.

The physical basis for this rank order was introduced in Section 2.2: the monthly displacement increments  $\delta(t)$  at all three sites are heavy-tailed (excess kurtosis +4.34 to +19.57). Variance, which squares deviations from the mean, is highly sensitive to extreme months—such as July 2017 at Baishuihe ( $\delta = 95$  mm)—whereas AR1, a Pearson correlation coefficient that depends only on the rank ordering of data points rather than their magnitude, is far less responsive to isolated outliers. This asymmetry explains why Variance consistently emerges as the strongest regime classifier across the three sites.

#### 4.4. Cross-site consistency and robustness checks

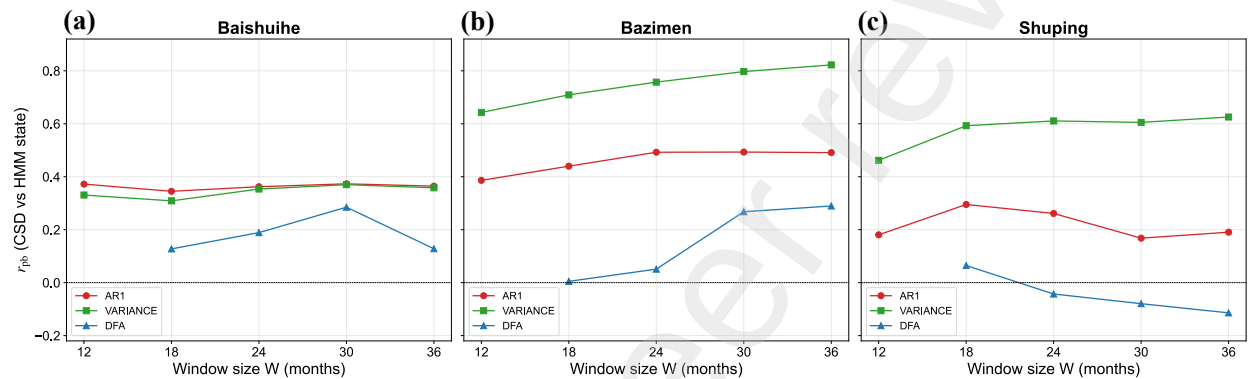
The analyses presented so far rest on a specific set of methodological choices: a sliding window of  $W = 24$  months, an event definition based on the 95th percentile, the primary monitoring points ZG93/ZG111/ZG86, and a single draw from the set of HMM random seeds. This section examines the sensitivity of the conclusions to each of these choices.

The window width  $W$  determines the temporal resolution of the CSD indicators and was set to 24 months on physical grounds (Section 3.2). To assess whether this choice materially affects the results, the full CSD–HMM point-biserial analysis was repeated at  $W \in \{12, 18, 24, 30, 36\}$  months. Across all three sites, five windows, and three indicators, every one of the resulting 45 point-biserial correlations remains positive (Fig. 4). The CSD–HMM association is not an artefact of a particular window choice. Variance yields the highest  $r_{pb}$  curve at every site and every window; its coefficient of variation across windows is  $CV < 0.5$ , qualifying as robust. AR1 meets the same criterion.

**Table 6**Steiger's  $Z$  test comparing CSD indicators as regime classifiers.

| Comparison       | Steiger's $Z$ (per site) |                 |         | Meta-analysis                             |
|------------------|--------------------------|-----------------|---------|-------------------------------------------|
|                  | Baishuihe                | Bazimen         | Shuping | Stouffer combined $Z$ ( $p$ , one-tailed) |
| Variance vs. AR1 | -0.04                    | <b>+4.99***</b> | -1.52   | <b>+2.88**</b> (0.002)                    |
| Variance vs. DFA | -0.30                    | +3.81           | +2.23   | +2.02 (0.022)                             |
| AR1 vs. DFA      | -0.25                    | -2.17           | +2.85   | +0.19 (0.425)                             |

Note. \*\*\*  $p < 0.001$ , \*\*  $p < 0.01$  (one-tailed). Steiger's  $Z$  tests whether two CSD indicators differ significantly in their point-biserial correlation with the same HMM regime labels (Steiger, 1980). Stouffer's method combines  $Z$ -statistics from the three sites for cross-site meta-analysis (Kocak, 2016).



**Figure 4:** CSD-HMM Window Sensitivity Analysis. Point-biserial correlation ( $r_{pb}$ ) between CSD indicators and HMM regime labels as a function of window size  $W \in \{12, 18, 24, 30, 36\}$  months. Three panels, one per site; AR1 (red), Variance (green), and DFA (blue) within each panel. The dashed line marks  $r_{pb} = 0$ .

DFA exhibits  $CV > 0.5$ , indicating window-dependent performance, consistent with its overall weakness throughout this study. The full per-window  $r_{pb}$  values are tabulated in Supplementary Table S7.

The HMM regime partition is insensitive to the random seed. Across five independent seeds, the Baum–Welch EM initialisation produces identical dormant-month proportions (fluctuation: 0.0 percentage points); the learned regime structure is deterministic, not a product of favourable initialisation (Supplementary Table S8). The choice of primary monitoring point is likewise not a source of systematic bias. Repeating the CSD–HMM point-biserial analysis with the auxiliary points (ZG118, ZG110, ZG85, ZG88) and the negative-control reference point (ZG91) yields sign agreement in eight of the nine primary–auxiliary comparisons, and a Steiger's  $Z$  test detects no significant difference between primary- and auxiliary-point results (Supplementary Table S9).

The sensitivity of the lead-time null result to the event-definition threshold has already been addressed in Section 4.1: across all 27 permutations of threshold (85th, 90th, 95th), site, and indicator, every permutation  $p$ -value exceeds 0.05 (Supplementary Table S5). The conclusion that CSD cannot provide event-level warning is robust to how generously the events are defined.

Finally, do the three landslides tell a consistent story? Kendall's  $W$  test of cross-site concordance addresses this question formally (Table 7). The rankings of the CSD–HMM point-biserial correlations across the three sites are significantly concordant for Variance (Kendall  $W = 0.133$ ,  $p < 0.001$ ): Variance is the best or joint-best indicator at every site. Neither AR1 nor DFA achieves significant cross-site concordance, indicating that their behaviour is more site-specific. Table 7 also reports the false-alarm characteristics of an 80th-percentile threshold warning strategy. The false-alarm rate varies widely by site and indicator: precision reaches 0.93 for AR1 at Baishuihe but drops to 0.18 for Variance at Shuping. These values are insensitive to the choice of warning threshold (70th, 80th, or 90th percentile; Supplementary Table S6). This cross-site heterogeneity reinforces the central message of the paper: CSD is not a one-size-fits-all event predictor, but a site-specific regime classifier.

**Table 7**

False alarm rate, precision, and cross-site Kendall concordance of CSD indicators as regime classifiers.

| Site      | 80th-percentile warning |      |           | Kendall $W$ |        |
|-----------|-------------------------|------|-----------|-------------|--------|
|           | Best indicator          | FAR  | Precision | $W$         | $p$    |
| Baishuihe | AR1                     | 0.07 | 0.93      | 0.133       | <0.001 |
| Bazimen   | Variance                | 0.39 | 0.61      |             |        |
| Shuping   | Variance                | 0.82 | 0.18      |             |        |

*Note.* FAR = false alarm rate (proportion of dormant months wrongly classified as active). Precision = proportion of warning months that coincide with HMM-active months. Kendall's  $W$ : coefficient of concordance across the three sites for each CSD indicator's  $r_{pb}$  ranking; only Variance achieves significant concordance ( $p < 0.001$ ).

The full set of sensitivity analyses thus supports the stability of the regime-classification findings across the examined range of methodological choices. The physical interpretation of these findings and their relationship to the CSD literature are examined in the Discussion.

## 5. Discussion

This study set out to determine whether critical slowing down (CSD) indicators could provide early warning for slow-moving reservoir landslides. After CSD computation across three sites and 17 years of monthly data, independent validation by an HMM, and seven sensitivity and robustness checks, the answer is more nuanced than the question.

CSD indicators do not provide statistically significant lead-time warning ahead of individual displacement acceleration events. All nine permutation  $p$ -values exceed 0.05, and this null result holds across 27 threshold-variant tests. The classic CSD early-warning successes in ecology—eutrophication of shallow lakes, desertification of semi-arid ecosystems—rest on hundreds to thousands of high-frequency observations (Scheffer et al., 2009). Landslide monthly displacement series, by contrast, are event-sparse and heavy-tailed. Under these conditions, individual-event prediction may not be a realistic expectation for CSD methodology: the sampling density may be insufficient to resolve a precursory signal above the noise floor. This result does not diminish the value of CSD indicators in the landslide context; rather, it clarifies where the methodology can and cannot contribute.

When the question shifts from predicting events to characterising dynamical regimes, CSD indicators show systematic and statistically significant discriminatory power. CSD indicators—Variance in particular—are consistently higher during HMM-active months than during HMM-dormant months, with six of the nine point-biserial correlations surviving Bonferroni correction. Critically, the HMM regime partition was learned from  $\delta(t)$  alone, with no access to the CSD indicators; two methods that share no theoretical foundation converge on the same temporal partition. We did not identify any previous study that applies an orthogonal HMM to validate CSD-based regime discrimination in landslide displacement data. Pu et al. (2015) applied CSD indicators descriptively to a single Three Gorges landslide but lacked an independent regime-validation tool. Nava et al. (2024) detected regime shifts in slow-moving landslides via VECC residual analysis—a cointegration-econometric framework unrelated to CSD. The CSD–HMM combination examined here follows a methodological path distinct from both: regime detection driven by physical theory rather than by model residuals, contributing a complementary perspective to the expanding research strand on regime identification in landslide early warning.

An important interpretative question concerns what Variance actually measures in this system. Because Variance squares deviations from the mean, it is highly responsive to large displacement increments; an increase in sliding-window Variance could in principle reflect either a genuine reduction in the system's resilience (the CSD mechanism) or simply the occurrence of large-amplitude deformation driven by intense external forcing. The present data do not allow a definitive separation of these two contributions. Several considerations, however, weigh against the interpretation that Variance is merely a proxy for forcing intensity. The hydrological cross-analysis (Supplementary Table S11) shows that HMM-active months are not systematically associated with stronger contemporaneous rainfall or larger water-level changes: at Bazimen and Shuping, antecedent rainfall during active months is significantly *lower* than during dormant months (Mann–Whitney  $p < 0.001$ ), and logistic regressions of HMM state on lagged rainfall and water-level change yield McFadden  $R^2$  values below 0.14. These results indicate that HMM-regime classification is not reducible to hydrological intensity, which in turn suggests that the CSD indicators associated with those

regimes are not functioning as simple proxies for external forcing. This interpretation is consistent with the established understanding that landslide kinematics in the Three Gorges region involve a non-linear, path-dependent coupling between hydrological forcing and displacement response (Du et al., 2013). For the operational purpose of regime classification—determining whether a landslide is in a dormant or active state—whether the CSD signal originates from reduced resilience or from the amplitude structure characteristic of the active regime is, in practical terms, immaterial: the indicators discriminate the two regimes with statistical confidence. Establishing a mechanistic link between CSD indicators and landslide failure physics would require coupling to hydrogeological models, which lies beyond the scope of the present phenomenological study.

Variance consistently outperforms AR1 as a regime classifier across all three sites, a finding that runs counter to the prevailing consensus in the CSD literature. Benchmark studies in ecology—most notably the systematic comparison by Dakos et al. (2012)—found AR1 to be more robust than Variance, with Variance occasionally declining rather than rising near critical transitions. The Steiger  $Z$  test yields the opposite rank order (Stouffer combined  $p = 0.002$ ). The heavy-tailed nature of landslide  $\delta(t)$  (excess kurtosis +4.34 to +19.57) likely contributes to this reversal: Variance squares deviations and thereby amplifies the contribution of extreme months—such as July 2017 at Baishuihe ( $\delta = 95$  mm)—whereas AR1, a rank-based correlation, is largely indifferent to them. The broader implication is that the relative performance of CSD indicators is system-dependent rather than universal. For over-damped, step-like systems with heavy-tailed increments—slow-moving landslides being a representative case—Variance may carry stronger regime-discriminating information than AR1. Morr & Boers (2024) (Morr and Boers, 2024) recently demonstrated that both indicators can fail under red-noise conditions, and Guttal, Raghavendra, Goel and Hoarau (2016) reported a similar decoupling in financial systems, where rising variance did not reflect a critical transition. The present results add a further dimension: even when the indicators are informative, their sensitivity ordering may depend on the statistical character of the system dynamics.

Four limitations should be addressed in future work. First, the sample is limited to three landslides. Although the Kendall  $W$  cross-site concordance (Variance  $p < 0.001$ ) and the weak cross-site displacement correlations ( $r = 0.22$ – $0.34$ ) provide statistical support for treating the three sites as independent lines of evidence, external validation on other reservoir landslides (e.g. Vajont, Italy; rainfall-triggered landslides in Japan) is needed. Second, the monthly sampling frequency is inherent to the available GPS monitoring data and may miss sub-monthly precursory signals. Higher-frequency data are not currently obtainable for the Three Gorges sites studied here. Third, the CSD and HMM analyses are retrospective: all data come from historical monitoring records, which is intrinsic to regime classification research, since the concept of a “regime” is itself only identifiable retrospectively. Fourth, CSD is a phenomenological framework: it infers critical dynamics from statistical precursor signals but does not provide a physical mechanism for landslide failure. Coupling CSD–HMM regime information with physically based hydrogeological models (e.g. rainfall infiltration–pore-pressure–displacement models) is a natural next step toward linking phenomenological signals to geotechnical understanding.

Two complementary directions merit pursuit. The first is integrating CSD–HMM regime information with existing hydrological-threshold early-warning systems. Current landslide warning in the Three Gorges region relies primarily on rainfall thresholds and reservoir-level drawdown rules, which define when a triggering condition may be present but cannot determine whether the system, in its current dynamical state, will respond. Regime information could provide an additional decision dimension and thereby potentially reduce false-alarm rates. The second direction is spatial extension. Sentinel-1 InSAR data provide surface-wide deformation measurements (Supplementary Fig. S8) that could extend the point-scale GPS analysis performed here to the spatial dynamics of the landslide body, making it possible to assess whether different sectors of the same landslide occupy the same dynamical regime.

## 6. Conclusion

This study examined whether critical slowing down (CSD) indicators could serve as early-warning tools for slow-moving reservoir landslides. The evidence supports two conclusions. First, CSD indicators cannot predict the timing of individual displacement acceleration events—a null result that holds across three independent landslides, nine permutation tests, and 27 threshold variants. Second, CSD indicators can characterise the underlying dynamical regime of a landslide, discriminating dormant from active months with statistical confidence that survives Bonferroni correction and is independently validated by a hidden Markov model trained on displacement data alone. The framework draws its validity from the convergence of two orthogonal theoretical approaches.

The most counter-intuitive finding is that Variance consistently outperforms AR1 as a regime indicator across all three sites, a rank order that reverses the consensus established by CSD research in ecology. This reversal may be related to the heavy-tailed nature of landslide monthly displacement increments—a property that is not specific to the Three Gorges sites but generic to step-like, hydrologically driven slow-moving landslides. For such systems, the relative performance of CSD indicators may reflect the underlying distribution of the data rather than a universal hierarchy. These results suggest that the choice of CSD indicator for regime classification in over-damped geohazard systems should be guided by the statistical properties of the target variable, with Variance as a natural candidate for systems characterised by heavy-tailed displacement increments.

## CRediT authorship contribution statement

**Zhou Fang:** Conceptualization, Methodology, Software, Validation, Formal Analysis, Investigation, Resources, Data Curation, Writing – Original Draft, Writing – Review & Editing, Visualization.

## Declaration of competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

The dataset is provided by National Cryosphere Desert Data Center (<http://www.ncdc.ac.cn>). The study uses 39 individual annual monitoring datasets spanning 2006–2025 for three landslides (Baishuihe, Bazimen, Shuping) in the Three Gorges Reservoir Area. Each dataset has a registered DOI; the complete list is provided in the accompanying file `data-sources.bib`. Processed datasets in Parquet format and all analysis code are deposited at Zenodo (<https://doi.org/10.5281/zenodo.21860470>) and will be made publicly available upon acceptance.

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## Declaration of generative AI

During the preparation of this work the author used Claude (Anthropic) for grammar checking and language polishing of the English prose. After using this tool, the author reviewed and edited all content and takes full responsibility for the content of the published article.

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